

COURSE DESCRIPTION			
Program	Chemical Engineering		
Course Title	Industrial Safety - Principles and Practices		
Course Code	2073	Semester	II
Type of Course	Theory	Contact Hours	60
Teaching Scheme (L: T:P:R)	4:0:0:0	Credits	4
CIE Marks	40	ESE Marks	60

Introduction

Industrial Safety - Principles and Practices introduces students to the fundamental concepts, tools, and behaviours required to prevent accidents and protect people, property, and the environment in industrial plants. The subject explains how different types of hazards (chemical, physical, biological, ergonomic, and psychological) arise in process industries and how their risks can be evaluated using toxicity data, exposure limits, and hazard indices. Building on this, it familiarises students with structured hazard analysis methods, the correct use of personal protective equipment, the basics of fire chemistry and fire protection systems, and essential operational controls such as permit-to-work, lockout/tagout, confined-space entry, and safety checklists so that they can participate effectively and safely in the operation of chemical processes.

Course Objectives

1	To develop fundamental awareness of industrial hazards and accident consequences.
2	To understand basic strategies used to eliminate, reduce, and control risks.
3	To become familiar with common engineering, administrative, and emergency protection measures used in industries.
4	To select and apply appropriate PPE and Fire Extinguishers, recognising PPE as the last line of defence.

Course Pre-requisite

Topic	Course code	Course Title	Semester
Basic understanding of force, pressure, heat, and electricity	2002B	Engineering Physics for Applied Electrical Technology and Computing	I
Basic understanding of acids, bases, reactions, and flammable substances	2003A	Chemistry for Engineering Practices	I
Basic knowledge of hazards in laboratories	2009B	Engineering Workshop Practice	I/II

Course Outcome

CO	Course Outcome	Blooms Taxonomy Level
CO1	Explain fundamental industrial safety concepts, identify and classify hazards, and describe basic fire, explosion, and toxicity principles.	Understand
CO2	Explain the hierarchy of controls and describe, at an introductory level, how elimination, substitution, and inherently safer approaches reduce industrial risk.	Understand
CO3	Describe the purpose and basic functioning of engineering and administrative control measures such as containment, detection, interlocks, permits, and emergency planning.	Understand
CO4	Select appropriate fire protection and personal protective measures by relating types of hazards to extinguishers and PPE and recognising their limitations.	Apply

CO-PO mapping

COURSE ARTICULATION MATRIX											
Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	-	-	3	-	-	3	-	-	-	-	2
CO2	-	-	3	-	-	3	-	-	-	-	2
CO3	-	-	3	-	-	3	-	-	-	-	2
CO4	-	-	3	-	-	3	-	-	-	-	2
Total	-	-	12	-	-	12	-	-	-	-	8
Correlation Level	-	-	3	-	-	3	-	-	-	-	2

Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - “-”

Teaching and Assessment Scheme

Teaching Scheme/Week				Self-learning (SS)/Week	Total Hours/Semester	Credits C	Examination Scheme						
							Theory			Practical			Total
L	T	P	S				CIE	ESE	Total	CIE	ESE	Total	
4	0	0	0	6	60	4	40	60	100	0	0	0	100

L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

Syllabus - Major Topics

Module	Title	Major Topics	Instructional Hours	
			L	T
Module I	Fundamentals of Industrial Safety	Industrial safety introduction - safety culture - individual responsibility - safety terminology (incident - accident - near-miss - risk - unsafe-act - unsafe-condition) - Industrial hazards classification - explosive limits - flammability - flash point - ignition temperature - toxicity - routes of entry - threshold exposure limits - chemistry of fire - fire triangle - fire tetrahedron - classification of fires - causes of fire - major industrial accidents - safety lessons.	15	0
Module II	Hazard Reduction and Inherently Safer Approaches	Hierarchy of controls - significance in risk reduction - source-level control - inherently safer design philosophy - substitution of hazardous materials - environmental and safety benefits - inventory reduction - safer process routes - advantages and limitations of source-level control - awareness of hazard analysis methods (PHA - What-If - JSA - HAZOP - FTA) - purpose and basic application of each method.	15	0
Module III	Engineering and Administrative Control Measures	Engineering controls - limitations of engineering controls - process containment - enclosures - barriers - segregation - ventilation - preventing unsafe pressure or temperature rise - fire and explosion prevention systems - hazard detection and alarm systems - automatic protection - administrative controls - limitations of administrative measures - permit-to-work systems - lockout/tagout intent - onsite and offsite emergency planning.	15	0
Module IV	Fire Protection and Personal Protective Equipment	Fire extinguishment - classification of fires - selection of suitable extinguishing media (water, foam, dry-chemical-powder, CO ₂ , special extinguishers) - limitations of extinguishers - PPE - PPE limitations - classification of PPE - PPE selection and use - awareness of modern wearable safety technologies.	15	0

Detailed Syllabus

Subtopic	Student Learning Outcome	Mode of delivery (L/T)	Mapped CO	Taxonomy Level	Instructional Hours
Module I					
Introduction to Industrial Safety	Define industrial safety. Explain the importance of industrial safety and the role of individuals in preventing accidents. Summarise the responsibilities of employees and the concept of safety culture.	L	CO1	Understand	1
Safety Terminologies	Define safety terminologies -Incident, Accident, Near Miss, Risk, Unsafe Act, Unsafe Condition, Onsite Emergency Plan, Offsite Emergency Plan	L	CO1	Remember	1
Industrial Hazard	Define hazard and explain the classifications of hazards.	L	CO1	Understand	2
Explosive limits	Define explosive limit. Differentiate between the lower explosive limit and the upper explosive limit. Explain the meaning of 'lower explosive limit'. and the 'upper explosive limit' with the example of LPG.	L	CO1	Understand	1
Flammability	Define flammability. Classify flammable liquids in terms of flash point.	L	CO1	Understand	1
Toxicity	Define toxicity and toxicity hazard. Summarise routes of entry - inhalation and skin ingestion. Describe the Threshold Limit Value (TLV) and the maximum safe exposure limit for employees.	L	CO1	Understand	1

Fire Chemistry	Describe the chemistry of fire. (Basic chemical nature of fire)	L	CO1	Understand	1
Fire Triangle / Tetrahedron	Explain elements of fire. Fire Triangle and Fire Tetrahedron	L	CO1	Understand	1
Classify fire	Classify fire according to combustible materials	L	CO1	Understand	1
Initiation of fire	List the causes of the initiation of fire. List the causes of electrical fires. Outline static electricity and static electricity hazards.	L	CO1	Understand	1
Major industrial accidents	Discuss the Bhopal disaster to identify its root causes and summarise the lessons learned to prevent recurrence in chemical process operations.	L	CO1	Understand	1
Major industrial accidents	Discuss the Flixborough disaster to identify its root causes and summarise the lessons learned to prevent recurrence in chemical process operations.	L	CO1	Understand	1
Major industrial accidents	Discuss the Texas City refinery explosion to identify its root causes and summarise the lessons learned to prevent recurrence in chemical process operations.	L	CO1	Understand	1
Major industrial accidents	Discuss the Seveso disaster to identify its root causes and summarise the lessons learned to prevent recurrence in chemical process operations.	L	CO1	Understand	1
Module II					
Hierarchy of controls	Explain the Hierarchy of Controls method developed by the National Institute for Occupational Safety and Health (NIOSH), USA.	L	CO2	Understand	2
Introduction to source-level control	Summarise the source-level control hierarchy. List the types of Hazards addressed by elimination and substitution. Describe how elimination and substitution reduce hazards with simple examples.	L	CO2	Understand	2
Basic Principles and ISD	Summarise the basic principles of hazard elimination. Outline the philosophy of inherently safer design – minimise, substitute, moderate, and simplify (Only the introduction of the concept)	L	CO2	Understand	2
Substitution	Describe the substitution of hazardous chemicals with safer alternatives using simple examples. Recognise at a basic level how safer alternatives promote reduced waste and safer environmental outcomes in line with the general intent of green chemistry. (Only the intent of green chemistry is to be introduced.)	L	CO2	Understand	2
Inventory Reduction and Safer Process Routes	Outline Inventory Reduction and Safer Process Routes. Explain in simple terms how reducing quantity or changing the process route lowers risk.	L	CO2	Understand	2
Pros & Cons	State the advantages and limitations of source-level control.	L	CO2	Understand	1
Hazard Analysis Methods	List and state the basic purpose of a Preliminary Hazard Analysis (PHA) in identifying potential hazards during the early stages of a process or project. (Detailed procedure is not required.)	L	CO2	Understand	1
Hazard Analysis Methods	List and state the basic purpose of What-If Analysis and Job Safety Analysis (JSA) in identifying operational and task-based hazards (Detailed procedure is not required)	L	CO2	Understand	1
Hazard Analysis Methods	List and state the basic purpose of a Hazard and Operability Study (HAZOP) in systematically identifying process deviations and possible risks. (Detailed procedure is not required.)	L	CO2	Understand	1
Hazard Analysis Methods	List and state the basic purpose of Fault Tree Analysis (FTA) in analysing the logical causes of system failures and accidents. (Detailed procedure is not required.)	L	CO2	Understand	1
Module III					
Introduction to Engineering Controls	Define engineering control and explain its purpose. Differentiate between source control and engineering control. Identify the limitations of engineering control.	L	CO3	Understand	1

Process containment and isolation	Summarise process containment and isolation. Describe at an introductory level how containment and physical barriers prevent release or exposure.	L	CO3	Understand	2
Ventilation and Exposure Control	Outline the ventilation and exposure control. State the role of ventilation in reducing employee exposure.	L	CO3	Understand	1
Pressure and Temperature Control Systems	Outline pressure and temperature control systems. Explain in simple terms why systems are provided to prevent unsafe increases in pressure or temperature.	L	CO3	Understand	1
Fire and Explosion Prevention Systems	Summarise fire and explosion prevention systems. Recognise the basic function of flame arresters' inerting and blanketing systems in preventing ignition or fire spread.	L	CO3	Understand	2
Hazard Detection and Monitoring Systems	Summarise common types of detectors (gas detectors, flame and smoke detectors) and state their warning function.	L	CO3	Understand	1
Engineering Safety Devices	Outline the idea of automatic safety devices such as emergency shutdowns and interlocks, and explain how they provide protection when unsafe conditions occur. (Only basic understanding of the safety instrumented systems)	L	CO3	Understand	1
Administrative Controls	Explain administrative controls and recognise their limitations compared to engineering measures.	L	CO3	Understand	1
Permit systems	Comprehend permit systems. Describe the purpose of work permits and situations where they are required (hot work, cold work, confined space and electrical work)	L	CO3	Understand	2
Lockout-tagout (LOTO)	Comprehend Lockout-tagout (LOTO). Explain the basic intent of lockout-tagout in preventing accidental energising.	L	CO3	Understand	1
Emergency response	Explain, in general terms, the purpose of onsite and offsite emergency planning and list the activities.	L	CO3	Understand	2
Module IV					
Fire extinguishing techniques	Explain fire-extinguishing techniques - cooling, smothering, starving, chain breaking	L	CO4	Understand	2
Fire extinguishing system classification	Classify fires based on the type of combustible material and relate them to suitable extinguishing media.	L	CO4	Understand	2
Water-type Extinguishers	Describe the general operating principle, suitability, and limitations of water-type extinguishers.	L	CO4	Understand	2
Foam-type extinguishers	Explain the basic principle of foam extinguishers and identify situations where they are used and their limitations.	L	CO4	Understand	1
Dry Chemical Powder (DCP) Extinguishers	Explain how dry chemical powder (DCP) extinguishers interrupt combustion. State their applications and limitations.	L	CO4	Understand	1
Carbon Dioxide (CO ₂) Fire Extinguishers	Explain the smothering action of CO ₂ extinguishers, typical uses and limitations.	L	CO4	Understand	1
Special Extinguishers	Recognise special extinguishers used for metal fires, their purpose and limitations.	L	CO4	Understand	1
Requirements of PPE	Explain why PPE is necessary and recognise its limitations as the last line of defence.	L	CO4	Understand	1
Types of PPE	Explain the classification of personal protective equipment: types of PPE, head protection, eye and face protection, Respiratory protection, and Body protection	L	CO4	Understand	2
Selection and Proper Use of PPE	Identify factors influencing PPE selection and describe correct usage practices. Select suitable PPE for a specified situation.	L	CO4	Understand	1
Wearable technologies in safety	State examples of modern wearable safety devices and their intended benefit.	L	CO4	Understand	1

Suggested Student Activities for Self-Learning

Week	Self-Learning description	Hours
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Module I		
1	Read textbook: Chemical Process Industry Safety. Observe safety practices on your campus/in your workshop. List at least five unsafe acts or unsafe conditions noticed in daily life. Prepare notes on unsafe acts & unsafe conditions. Watch an introductory video on "Importance of Industrial Safety"	6
2	Collect real or news examples of accidents and near misses. Try to categorise them. Prepare a short note differentiating risk and hazard. Explore the OSHA website for basic industrial safety. Prepare study notes.	6
3	Identify hazards in home, road, laboratory or construction areas. Classify them (mechanical, electrical, chemical, etc.). Watch an online video on hazard identification. Prepare study notes.	6
4	Read about LPG safety & hazard communication symbols. Observe LPG safety practices at home or hostel. List precautions taken to prevent leaks and ignition. Recognise hazard labels from household chemicals. Identify warnings and protective advice. Prepare study notes.	6
Module II		
5	Study inherently safer design principles. Find examples where safer materials replaced dangerous ones (fuels, cleaners, etc.). Prepare comparison notes. Watch a short video on the substitution principle in safety.	6
6	Read about minimisation and moderation concepts. Think of situations where using smaller quantities improves safety. Discuss in groups where smaller quantities improve safety and prepare a group discussion summary. Explore chemical safety data sheets (SDS) online. Prepare study notes.	6
7	Read and self-study of hazard analysis techniques (PHA, What-If, HAZOP basics). Prepare a simple comparison table. Summarise learning and prepare study notes.	6
8	Study safety barriers and engineering controls. Identify barriers, guards, or enclosures around you. Explain what risk they reduce. Explore a short industry safety case video. Summarise learning and prepare study notes. Prepare for an assignment covering CO1 and CO2.	6
Module III		
9	Study various engineering control systems in a crude storage tank farm. Search online images of tank farm safety systems. Summarise learning and prepare study notes.	6
10	Self-study of permit-to-work systems using industry guidelines. Download the sample PTW format online. Prepare a summary note and identify key components.	6
11	Study Lockout-Tagout (LOTO) & confined space safety. Review of lockout-tagout and confined space safety videos. Prepare a checklist for confined space entry. Summarise learning and prepare study notes.	6
12	Study of accident investigation reports and safety checklists. Prepare a basic flowchart of investigation steps. Summarise learning and prepare study notes.	6
Module IV		
13	Self-study on types, standards, and limitations of PPE. Identify and list the PPE to be used in the lab/workshop. Summarise learning and prepare study notes.	6
14	Study of PPE selection charts and exposure limits. Summarise learning and prepare study notes.	6
15	Review of occupational exposure monitoring methods. Study human factors & behavioural safety concepts. Identify wearable safety devices used in industry. Summarise learning and prepare study notes. Prepare for an assignment covering CO3 and CO4	6
Total Self-Learning hours		90

Learning Resources

TEXT BOOK

Sl. No.	Title	Author	Publication
1	Chemical Process Industry Safety	K. S. N. Raju	McGraw-Hill Education India
2	Industrial Safety, Health and Environmental Management Systems	R. K. Jain & Sunil S. Rao	Khanna
3	Industrial Safety Management: Hazard Identification and Risk Control	L. M. Deshmukh	McGraw-Hill Education India

REFERENCE			
Sl. No.	Title	Author	Publication
1	Safety in the Chemical Laboratory and Industry -A Practical Guide	Muhammad Raza Shah	Elsevier
2	Chemical Process Safety: Fundamentals with Applications	Daniel A. Crowl & Joseph F. Louvar	O'Reilly

WEB RESOURCES		
Sl. No.	URL	Modules Covered
1	https://www.nptelprep.in/courses/110105094	I,II,III
2	https://onlinecourses.swayam2.ac.in/nou26_ge33/preview	I,II,IV

Assessment Methodology - Theory

Mode	Attendance	CA1	CA2	CA3	CA4	CA5	CA6	ESE
		Self-learning activities (Module 1)	Self-learning activities (Module 2)	Self-learning activities (Module 3)	Self-learning activities (Module 4)	Series Exam (2 modules)	Series Exam (2 modules)	Written Exam (theory)
Duration						2 hours	2 hours	2.5 hours
Exam mark		15	15	15	15	50	50	60
Converted to						20	20	
Marks	5	15				20		60
Total	40							60
Tentative schedule	End of semester	By 4th week	By 7th week	By 11th week	By 14th week	8th week	15th week	End of semester

Series Examination Pattern- Theory

Time	Module	Part A	Marks	Part B	Marks	Part C (With choice)	Marks	Total
2 hours	Any one module	2	1	3	3	2	7	25
	Any one module	2	1	3	3	2	7	25
	Number	4		6		4		50

End Semester Examination Pattern of Question Paper - Theory

Duration	Module	TYPE OF QUESTIONS						Total Marks
		Part A (2 Questions from each module)		Part B 2 out of 3 questions from each module)		Part C (1 set questions with choice from each module)		
		No. of Questions	Marks (1 mark each)	No. of Questions	Marks (3 marks each)	No. of Questions	Marks (7 marks each)	
2.5 hours	1	2	2	3	6	1 set	7	15
	2	2	2	3	6	1 set	7	15
	3	2	2	3	6	1 set	7	15
	4	2	2	3	6	1 set	7	15
		8	8	12	24	4 sets	28	60